(SSUET/QR/111)

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT
BS (Computer Science)

COURSE INFORMATION SHEET

Session:	Spring 2024
Course Title:	Multivariate Calculus
Course Code:	MS-208
Credit Hours:	3
Semester:	4^{th}
Pre-Requisites:	Calculus
Instructor Name:	Ms. Fozia Iqbal
Email and Contact Information:	foiqbal@ssuet.edu.pk
WhatsApp Group	MultiCal CS Spring 2024
Office Hours:	*0900 hours to 1700 hours (Monday to Friday)*
Mode of Teaching:	Synchronous

COURSE OBJECTIVE:

The course aims to impart a deep understanding of advanced mathematical concepts including functions of several variables, differentiation and integration, vector calculus, and transform methods to the students of computer science. Students will learn techniques such as limit computation, partial differentiation, gradient vectors, and integral calculus, along with applications in optimization and function approximation. Additionally, the course covers vector fields, fundamental theorems such as Green's theorem and Stoke's theorem, and transforms including Laplace transforms, and Z-transforms, preparing students for diverse applications in engineering, physics, and applied mathematics.

COURSE OUTLINE:

Functions of Several Variables, limit, continuity, and Partial Differentiation. Mixed derivative theorem, increment theorem, chain rule, implicit differentiation theorem, gradient vector, tangent plane, and normal lines, linearization, extreme values, Lagrange multipliers, Taylors formula, Multiple Integrals, Line and Surface, Cylindrical and spherical coordinates, vector fields, Green's and Stoke's Theorem., divergence theorem, vector algebra, Fourier Transform; Laplace Transform, Z-Transform.

COURSE LEARNING OUTCOMES (CLOs) and its mapping with Program Learning Outcomes(PLOs):

CLO No.	Course Learning Outcomes (CLOs)	PLOs	Bloom's Taxonomy
1	**Understand** the concept of the functions of Several Variables, limit, continuity, and Partial Differentiation	PLO_1 (Academic Education)	C2 (Understanding)
2	**Applying** the concept of Vector algebra and the equations of Line and Surface	PLO_1 (Academic Education)	C3 (Applying)
3	**Applying** the concept of Fourier Transform; Laplace Transform, Z-transform.	PLO_2 (Knowledge for solving computing problems)	C3 (Applying)

COMPLEX ENGINEERING PROBLEM:

Complex Engineering Problem Details	Included: No

RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:

Assessment Tools	CLO-1 (38)	CLO-2 (34)	CLO-3 (28)	Total (100)
Quizzes	3 (8%)	3 (9%)	4 (14%)	10
Assignments	3 (8%)	3 (9%)	4 (14%)	10
Midterm Exam	20 (53%)	10 (29%)	--	30
Final Exam	12 (31%)	18 (53%)	20 (72%)	50

GRADING POLICY:

Assessment Tools	Percentage
Quizzes	10%
Assignments	10%
Midterm Exam	30%
Final Exam	50%
TOTAL	**100%**

RECOMMENDED BOOK:

- Maurice D. Weir, Joel Hass, Frank R. Giordano, ***Thomas' Calculus***, 14th edition, Pearson Prentice Hall., (2017). ISBN 13: 978-0-13-443898-6, ISBN 10: 0-13-443898-1
  https://www.engbookspdf.net/thomas-calculus-14th-edition-pdf/

- Erwin Kreyszig, Advanced Engineering Mathematics, (10th edition), John Wiley & Sons, Inc. (2011) ISBN 978-0-470-45836-5
  http://www.wiley.com/college/kreyszig

REFERENCE BOOKS:

- Anton, Bivens, Davis, Calculus, 11th edition, John Willey & Sons, Inc., (2016). ISBN-13: 978-8126556403, ISBN-10: 9788126556403
- Earl William Swokoski, ***Calculus***, 5th edition, PWS-KENT Pub. Co., (1991). ISBN 13: 978-0534926489, ISBN 10 0534435386
- Hughes-Hallett, Gleason, McCallum, et al, *Calculus Single and Multivariable*, 3rd Edition. John Wiley & Sons, Inc. 2002.
- H. K. Dass, *Advanced Engineering Mathematics,* S. Chand & Company Ltd

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LECTURE PLAN

Course Title: **Multivariate Calculus**
Course Code: **MS-208**

Week No.	Week Dates	Topics	Required Reading	Key Date
1	12-03-2024 to 15-03-2024	Function. Function of several variable. Domain and range. Boundary and interior points. Bounded and unbounded region. level curves and surfaces.	Chap#14 Exe. 14.1 1- 15 Page # 798	
2	18-03-2024 to 22-03-2024	Limit and continuity Properties of limit of function Continuous function	Chap#14 Exe. 14.2 1-5,13-16,27-34 Page # 806-807	
3	25-03-2024 to 29-03-2024	Partial derivative, continuity, Second order partial derivative, The derivative theorem Chain rule, Implicit differentiation theorem	Exe. 14.3 23-32, 41-50 Page # 818 Exe. 14.4 1-6, 25-30 Page # 828	Assignment # 1
4	01-04-2024 to 05-04-2024	Directional derivative, Gradient vector	Exe. 14.5 1-6, 11-13 Page # 838	
5	08-04-2024 to 12-04-2024	Tangent and normal lines, Linearization, Total differentiation	Exe. 14.6 1-6, 27-30 Page # 846	Quiz # 1
6	15-04-2024 to 19-04-2024	Extreme values, Saddle point, Constrained maxima and minima Lagrange multipliers	Exe. 14.7 1-5 Page # 856 Exe. 14.8 1-5 Page # 865	
7	22-04-2024 to 26-04-2024	Taylor formula, Double integral over rectangle. Surface area as a double integral Triple integral	Exe. 14.9 1-5 Page # 872 Exe. 15.1 17-21 Page # 887 Exe. 15.5 7-8 Page # 887	Assignment # 2

8	29-04-2024 to 03-05-2024	Vector algebra Analytics equation for line and plane Cylindrical and spherical coordinates	Exe. 12.5 1-8, 21-25 Page # 735 Article 15.7 Page # 934	
9	**Midterm Examination (06-05-2024 to 11-05-2024)**			
10	13-05-2024 to 17-05-2024	Parametric equations for 3D geometry. Gradient, Divergence and curl.	Exe. 16.4 1-4 Page # 996 Exe. 16.8 1-4 Page # 1042 Exe. 16.7 1-4 Page # 1029	
11	20-05-2024 to 24-05-2024	Vector fields, Line integrals Green's theorem.	Exe. 16.2 1-4, 7-10, 19 Page # 735 Exe. 16.4 7-10 Page # 996	Quiz # 2
12	27-05-2024 to 31-05-2024	Surface integral Stokes theorem	Exe. 16.6 1-6 Page # 1016 Exe. 16.7 7-8 Page # 1029	
13	03-06-2024 to 07-05-2024	Divergence theorem	Exe. 16.8 9-10 Page # 1042	
14	10-06-2024 to 14-06-2024	Fourier Integrals, Fourier Sine and Cosine Integrals	*Exe. 11.7 2-4, 7-10, 16-17 Page # 517	Assignment #3
15	17-06-2024 to 21-06-2024	Fourier transforms	*Exe. 11.9 1-5, 7, 10 Page # 533	
16	24-06-2024 to 28-06-2024	Laplace transforms of elementary functions, inverse transform	*Exe. 6.1 1-8, 25-30 Page # 210-211	Quiz # 3
17	01-07-2024 to 05-07-2024	Laplace transform, s-shifting, unit step functions, t-shifting	*Exe. 6.1 33-38 Page # 211 *Exe. 6.3	

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			2-14 Page # 223	
18	08-07-2024 to 12-07-2024	*Z*-transformation and its properties	Notes will be provided	
Final Examination **(15-07-2024 to 26-07-2024)**				

Thomas' Calculus, 14^{th} edition, Pearson Prentice Hall., (2017)
* *Advanced Engineering Mathematics*, (10th edition), John Wiley & Sons, Inc. (2011)

Name & Signature (Course Instructor): Fozia Iqbal __________ Date: 3^{rd} March'2024

Name: & Signature (Head of Department): __________________ Date: __________________